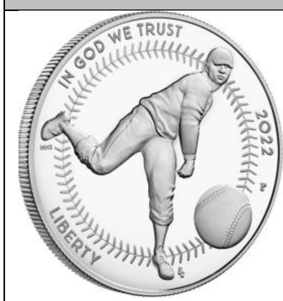
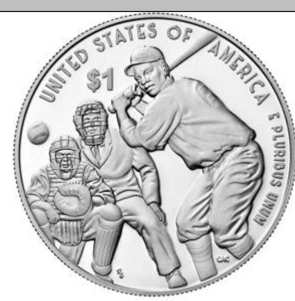
Grade 6 Accelerated
Unit 15 Assessment

Name____________________
Date_________________
Period__________

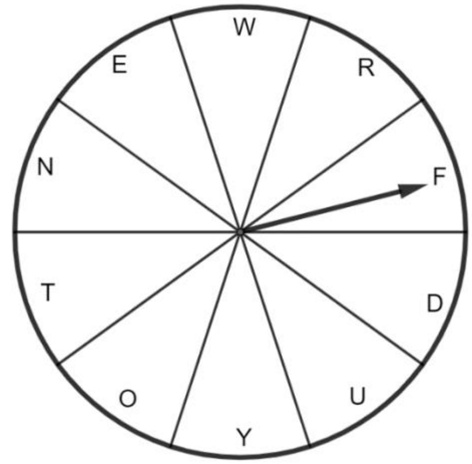
1. In 2022, the Negro Leagues Baseball Silver Dollar was released by the U.S. Mint as part of the Negro Leagues Baseball Commemorative Coin Program. The faces of the coin are shown.

2022 Negro Leagues Commemorative Silver Dollar	
Pitcher	**Batter**
	

For each simple experiment in the table, determine the sample space.

	Simple Experiment	Sample Space
a.	Flipping the Negro Leagues Silver Dollar	
b.	Rolling a fair $17-$ sided number cube 	

2. In a game, players spin a fair spinner to determine a letter that they must use. The spinner has 7 consonants and 3 vowels on it.



Find the theoretical probability of each event. Write the probability as a percentage.

	Event	Theoretical Probability
a.	Spinner lands on the letter "T".	
b.	Spinner lands on a consonant.	

3. The probabilities of four events are shown.

$$75\% \qquad 9 \; out \; of \; 10 \qquad \frac{8}{11} \qquad 0.77$$

List the probabilities in order from least likely to most likely to occur.

Least to Most Likely to Occur	

4. For each context, place an **X** in the box for all of the simulations that could be used to simulate the probability.

	Simulations		
	A deck of 16 cards with 16 different letters.	A fair spinner with 7 equal sections where each is a different color.	A 10-sided die with the numbers 1 through 10.
A baseball player with an on-base percentage of 70%.	☐	☐	☐
A basketball player making $\frac{5}{7}$ free throws.	☐	☐	☐
A soccer goalie who stops $\frac{1}{2}$ of shots on the goal.	☐	☐	☐
A hockey player with a shooting percentage of 25%.	☐	☐	☐

5. Complete the statement.

To use the

- ○ 10-sided die
- ○ deck of 16 cards
- ○ fair spinner with 7 equal sections

to simulate the

- ○ soccer
- ○ hockey
- ○ baseball
- ○ basketball

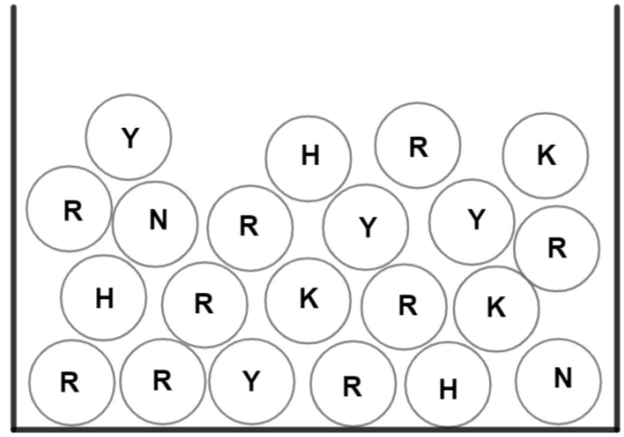
context, 5

- ○ cards
- ○ numbers
- ○ equal sections

should be assigned to

- ○ being on base.
- ○ making a free throw.
- ○ stopping a shot on goal.
- ○ shooting a goal in hockey.

6. A jar with 21 marbles is shown.



Determine each theoretical probability. Write the answer in fraction form.

	Event	Theoretical Probability
a.	Drawing a marble with the letter R or Y from the jar	
b.	Drawing a marble with a letter other than H from the jar	
c.	Drawing a marble with the letter N from the jar	

d. Complete the statements.

The event that is less likely to occur is

- ○ drawing a marble with the letter N from the jar.
- ○ drawing a marble with the letter R or K from the jar.
- ○ drawing a marble with a letter other than H from the jar.

The event that is most likely to occur is

- ○ drawing a marble with the letter N from the jar.
- ○ drawing a marble with the letter R or K from the jar.
- ○ drawing a marble with a letter other than H from the jar.

The probability of the less likely event is closer to

- ○ 0.
- ○ 1.

7. Aaliyah and Xavier did a computer simulation for drawing a marble with the letter R from the jar of 21 marbles. Each used a spinner with 21 equal sections where 3 sections are labeled H, 9 sections are labeled R, 3 sections are labeled K, 4 sections are labeled Y, and 2 sections are labeled N. The number of spins and the results are shown in the table.

	Aaliyah	**Xavier**
Spins	42	$21,000$
Results	$\frac{10}{42}$	$\frac{8982}{21000}$

The number of spins in
- ○ Aaliyah's
- ○ Xavier's

simulation results in a(n)

- ○ experimental
- ○ theoretical

probability that is closer to the

- ○ experimental
- ○ theoretical

probability.